RESEARCH STATEMENT

AI for Healthcare Enterprise Cybersecurity

Raj Patel · PhD Student, Department of Computer Science, The University of Alabama · rpatel38@ua.edu · rajpatel-ua.github.io

How healthcare enterprises can use machine learning to detect cyberattacks across wearables, medical-device gateways and enterprise networks, without exposing sensitive telemetry and without trusting a learning pipeline that may itself be compromised.

00 FOUNDATIONS

The work this builds on

Three threads from my published and accepted work lead directly into this direction.

THREAT MODEL

Secure MLOps

A lifecycle taxonomy of attacks and mitigations for machine-learning pipelines, which identified gaps in realistic defense evaluation and privacy protection.

First author · ACM TAISAP 2026

EVALUATION RIGOR

Leakage-resistant benchmarking

A benchmark of spiking neural networks for intrusion detection that showed how much results depend on separating training and evaluation data.

First author · IEEE CogMI 2026 · Best Student Paper Award

SUPERVISED AUTONOMY

Agentic cyber defense

Multi-agent intrusion response that keeps analysts in the approval loop, and defenses for the agents themselves.

Agentra and AgenticCyOps · ICMLA 2026; AIDC '26

01 DIRECTION

Research direction

I propose to investigate how healthcare enterprises can use machine learning to detect cyberattacks across wearables, medical device gateways, and enterprise networks while limiting the exposure of sensitive telemetry. My first-author *Towards Secure MLOps* survey maps attacks and mitigations across the machine learning lifecycle and identifies gaps in realistic defense evaluation and privacy protection [1]. I will extend that analysis into federated intrusion detection, where the learning pipeline itself must withstand compromise. My SNN intrusion-detection work, which received the Best Student Paper Award at IEEE CogMI 2026, also motivates careful separation of training and evaluation data. The initial scope is network security monitoring and the interaction of learning, deployment, and monitoring controls under attack.

02 QUESTION

Research question and threat model

Can update-integrity checks and drift-aware local detection improve attack resistance in federated healthcare intrusion detection at a fixed false-alert budget, without centralizing raw telemetry?

I hypothesize that authenticated update provenance and validation against reserved reference traffic will reduce poisoning success relative to standard federated averaging [2]. The threat model will include compromised clients submitting poisoned updates, adversaries modifying observable traffic to evade detection, and a curious aggregation service. Authentication identifies an update's source; authenticated clients may still be malicious.

ADVERSARY 1

Compromised clients

Gateways or sites that submit poisoned model updates, even when properly authenticated.

ADVERSARY 2

Evasive attackers

Adversaries who modify observable traffic so the detector misses them.

ADVERSARY 3

Curious aggregator

An aggregation service that follows the protocol but tries to learn from what it sees.

03 METHODOLOGY

Preliminary methodology

I will construct a reproducible testbed representing wearable or medical-device traffic, gateways, and enterprise services. Public datasets and controlled traffic generation will establish an initial benchmark; access to clinical systems is not assumed. Gateways or sites will train locally and exchange model updates. Device diversity, unequal client data, changing traffic, and intermittent connectivity will be varied explicitly. Training and evaluation will be separated by time, device, and site where the data support these splits, with preprocessing fitted only to training data.

The baseline systems will include local training, standard federated averaging, and a centralized reference under equivalent data and compute budgets. The proposed pipeline will check client identity, model version, and update quality before promotion. Ablations will remove update validation or drift adaptation. Controlled poisoning and evasion trials will vary attacker access, malicious-client fraction, and attack strength; held-out attacks will assess generalization. I will distinguish controls that inspect individual updates from configurations using secure aggregation and test the evidence each can access. Differential privacy experiments will specify the protected unit and report privacy accounting and utility loss.

I will evaluate attack success, detection quality at matched false-alert rates, detection latency, communication volume, and gateway inference cost. Repeated trials will report uncertainty and results by device and site. Initial experiments will produce alerts; later containment studies will require operator approval and measure disruption to clinical services. I will map access, audit, integrity, and transmission controls to HIPAA Security Rule safeguards [3], documenting their technical coverage and trust assumptions.

TESTBED	BASELINES	ABLATIONS
Wearable and medical-device traffic, gateways, enterprise services	**Local training, federated averaging, centralized reference**	**Without update validation; without drift adaptation**
ATTACKS	METRICS	COMPLIANCE
Poisoning and evasion, varying access, fraction and strength	**Attack success, detection at matched false-alert rates, latency, cost**	**Controls mapped to HIPAA Security Rule safeguards**

04 CONTRIBUTION

Anticipated scholarly contribution

The study will produce a healthcare-oriented lifecycle threat model, a reproducible benchmark, and an evaluated federated defense prototype. Its contribution will be an explanation of when heterogeneous clients and compromised updates cause operational security failures, which controls interrupt those failures, and how privacy protections change detection utility and cost. This will connect the survey's security taxonomy to measurable requirements for healthcare cyber defense.

- Healthcare lifecycle threat model
- Reproducible benchmark
- Evaluated federated defense prototype

05 REFERENCES

References

[1] R. Patel et al. "Towards Secure MLOps: Surveying Attacks, Mitigation Strategies, and Research Challenges." *ACM Transactions on AI Security and Privacy*, 2026. doi:10.1145/3796728

[2] H. B. McMahan et al. "Communication-Efficient Learning of Deep Networks from Decentralized Data." *AISTATS*, PMLR 54, pp. 1273–1282, 2017.

[3] U.S. Department of Health and Human Services, Office for Civil Rights. "Summary of the HIPAA Security Rule." hhs.gov